F78PA Probability & Statistics A

Draft Class test (Malaysia Campus)

Thursday, October 9th, 11am-12pm

1. In a state election, all voters live in three cities in the state with voter percentages in each city given as follows:

   - City A: 20%
   - City B: 30%
   - City C: 50%

   Further, the following information is given about an election candidate W:

   - 30% of voters in City A will vote for W,
   - 50% of voters in City B will vote for W, and
   - 40% of voters in City C will vote for W.

   (a) Calculate the probability that a randomly selected voter from the state will vote for W in the election.

   (2 marks)

   (b) Given that a voter voted for W, find the probability that the voter was from City A.

   (2 marks)

   (c) Are the events "Voter voted for W" and "Voter is from City A" independent? Justify your answer.

   (3 marks)

   (d) Given that a voter voted for candidate W and is not from City A, find the probability that the voter is from City B.

   (3 marks)

   [Total 10 marks]

2. The waiting time (in minutes) for a customer in a service queue, T, has the following probability density function

$$f_T(t) = \begin{cases} c\,t^2, & \text{for } 0 \le t \le 2 \\ 0, & \text{otherwise.} \end{cases}$$

   where c is a constant.

   (a) State the conditions required for a function f_T to be a valid probability density function and use these conditions to show that $c = 3/8$. Show all your steps.

   (2 marks)

   (b) Determine the cumulative distribution function F_T. Sketch the graph of $F_T(t)$ for $-\infty < t < \infty$.

   (4 marks)

   (c) Calculate the probability that a customer waits between 0.5 and 1.5 minutes in the queue.

   (2 marks)

   (d) 75% of the customers will wait less than w minutes for service. Find w.

   (2 marks)

   [Total 10 marks]